



THE SCOTS COLLEGE

2012

HSC Task 3 – In Class Test
13th June 2012

Mathematics Extension 2

General Instructions

- Working time – 45 minutes
- Write using black or blue pen
Black pen is preferred
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- Show all necessary working

Total marks – 30

Learning Intentions:

Integration and Volumes

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

QUESTION ONE (15 MARKS) Use a SEPARATE writing booklet.

a) $\int \sin^3 \theta \, d\theta$ (2)

b) $\int x e^x \, dx$ (2)

c) By using the substitution $t = \tan \frac{\theta}{2}$, evaluate in exact form (3)

$$\int_0^{\frac{\pi}{3}} \frac{d\theta}{5 - 4\cos\theta}$$

d) $\int_{-2}^2 (x\sqrt{4-x^2} - \sqrt{4-x^2}) \, dx$ (3)

e) Given that

$$I_n = \int_1^e (1 - \ln x)^n \, dx, (n = 1, 2, 3, \dots)$$

i. Show that $I_n = -1 + nI_{n-1}$ (3)

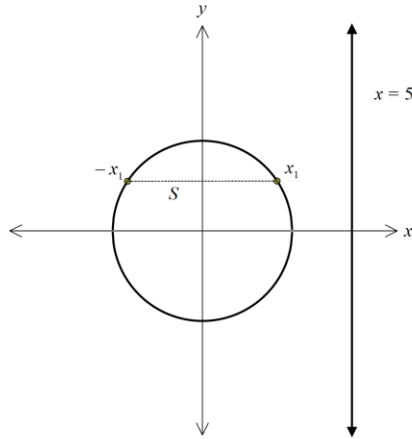
ii. Hence evaluate (2)

$$\int_1^e (1 - \ln x)^3 \, dx$$

End of Question 1

QUESTION TWO (15 MARKS) Use a SEPARATE writing booklet.

- a)** The circle $x^2 + y^2 = 9$ is rotated about the line $x = 5$ to form a ring. When the circle is rotated, the line segment S at height y sweeps out an annulus. The coordinates of the end-points of S are x_1 and $-x_1$, where $x_1 = \sqrt{9 - y^2}$.



- i.* Show that the area of the annulus is equal to (2)

$$20\pi\sqrt{9 - y^2}$$

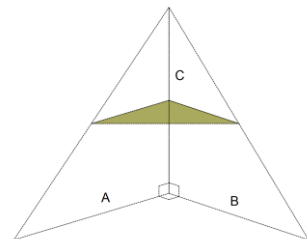
- ii.* Hence find the volume of the ring. (2)

- b)** *i.* Sketch region bounded by the curve $y = 4x - x^2$, $x = 1$, $y = x$ and the y -axis. (1)

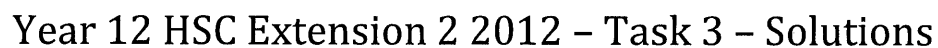
- ii.* This region is rotated about the y -axis. Use the method of cylindrical shells to calculate the volume of the solid created. (3)

- c)** A particular solid has as its base the region bounded by the hyperbola $x^2 - y^2 = 1$ and the line $x = 2$. Cross-sections perpendicular to this base and the x -axis are semi-circles whose diameter are in the base. Find the volume of the solid. (3)

- d)** A tetrahedron has three mutually perpendicular faces and three mutually perpendicular edges of length A, B and C . By slicing parallel to the base and summing such slices, confirm the formula for the volume of a tetrahedron is $V = \frac{ABC}{6}$. (4)

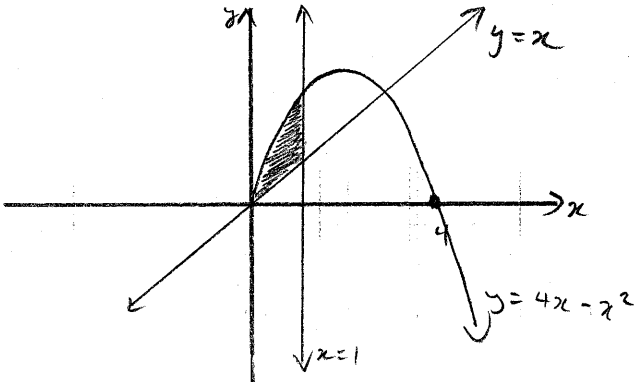


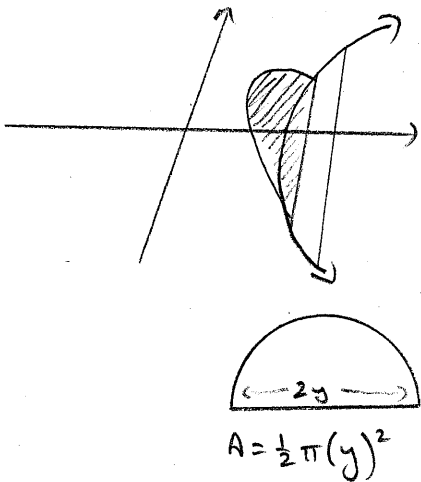
End of Paper



Q1	a)	$\int \sin^3 \theta \, d\theta = \int \sin \theta (1 - \cos^2 \theta) \, d\theta$ $= \int (\sin \theta - \sin \theta \cos^2 \theta) \, d\theta$ $= -\cos \theta + \frac{1}{3} \cos^3 \theta + C$		1 Method
	b)	$\int x e^x \, dx = e^x (x - 1) + \text{constant}$		1 Solution
		Possible intermediate steps: $\int e^x x \, dx$ For the integrand $e^x x$, integrate by parts, $\int f \, dg = fg - \int g \, df$, where $f = x, \quad dg = e^x dx,$ $df = dx, \quad g = e^x:$ $= e^x x - \int e^x \, dx$ The integral of e^x is e^x: $= e^x x - e^x + \text{constant}$ Which is equal to: $= e^x (x - 1) + \text{constant}$		1 for by parts 1 for solution

c)	$\int_0^{\frac{\pi}{3}} \frac{1}{5-4\cos(x)} dx = \frac{2\pi}{9} \approx 0.698132$		
	Possible intermediate steps: $\int \frac{1}{5-4\cos(x)} dx$ For the integrand $\frac{1}{5-4\cos(x)}$, substitute $u = \tan\left(\frac{x}{2}\right)$ and $du = \frac{1}{2} \sec^2\left(\frac{x}{2}\right) dx.$ Then transform the integrand using the substitutions $\sin(x) = \frac{2u}{u^2+1}, \cos(x) = \frac{1-u^2}{u^2+1} \text{ and } dx = \frac{2 du}{u^2+1};$ $= \int \frac{2}{(u^2+1)\left(5-\frac{4(1-u^2)}{u^2+1}\right)} du$ Simplify the integrand $\frac{2}{(u^2+1)\left(5-\frac{4(1-u^2)}{u^2+1}\right)}$ to get $\frac{2}{9u^2+1}$: $= \int \frac{2}{9u^2+1} du$ Factor out constants: $= 2 \int \frac{1}{9u^2+1} du$ The integral of $\frac{1}{9u^2+1}$ is $\frac{1}{3} \tan^{-1}(3u)$: $= \frac{2}{3} \tan^{-1}(3u) + \text{constant}$ Substitute back for $u = \tan\left(\frac{x}{2}\right)$: $= \frac{2}{3} \tan^{-1}\left(3 \tan\left(\frac{x}{2}\right)\right) + \text{constant}$	1 mark Set up 1 mark correct substitution	
	$\int_0^{\frac{\pi}{3}} \frac{1}{5-4\cos(x)} dx = \frac{2\pi}{9} \approx 0.698132$	1 mark solution	
d)	$\int_{-2}^2 (x\sqrt{4-x^2} - \sqrt{4-x^2}) dx$ $\underbrace{\int_{-2}^2 x\sqrt{4-x^2} dx}_{\text{odd fn}} - \underbrace{\int_{-2}^2 \sqrt{4-x^2} dx}_{\text{semi-circle radius 2}}$ $0 - \frac{1}{2} \pi r^2$ $= -\frac{1}{2} \pi \times 4 = -2\pi$	1 method 1 recognise odd & semi-circle 1 solution	

	e) i) $I_n = \int_1^e (1 - \ln x)^n dx, (n = 1, 2, 3, \dots)$ $= \left[x(1 - \ln x)^n \right]_1^e - \int_1^e n x (1 - \ln x)^{n-1} \cdot \left(-\frac{1}{x}\right) dx$ $= [0 - (1-0)^n] + n \int_1^e (1 - \ln x)^{n-1} dx$ $I_n = -1 + n I_{n-1}$	1 method	1 progress	1 solution
	ii) $\int_1^e (1 - \ln x)^3 dx = -1 + 3 I_2$ $= -1 + 3[-1 + 2 I_1]$ $= -1 + 3[-1 + 2 \int_1^e (1 - \ln x) dx]$ $= -1 - 3 + 6[-1 + I_0]$ $= -10 + 6 \int_1^e 1 dx$ $= -10 + 6[x]_1^e$ $= -10 + 6e - 6 = 6e - 16$	1 method		1 solution
Q2	a) i) $A = \pi R^2 - \pi r^2$ where $R = 5 + x_1$ $r = 5 - x_1$ $A = \pi(5 + x_1)^2 - \pi(5 - x_1)^2$ $= \pi(5 + x_1 - 5 + x_1)(5 + x_1 + 5 - x_1)$ $= 20\pi x_1 = 20\pi \sqrt{9 - y^2}$	1 set up		1 solution
	ii) $V = 2 \int_0^3 20\pi \sqrt{9 - y^2} dy$ $= 40\pi \int_0^3 \underbrace{\sqrt{9 - y^2}}_{\text{quarter circle}} dy$ $V = 40\pi \left(\frac{1}{4} \pi r^2 \right)_{r=3}$ $= 90\pi^2 \text{ u}^3$	1 correct integral		1 solution
	b) i) 	1 correct region		

	b) ii) $\lim_{\delta x \rightarrow 0} \sum_{x=0}^1 2\pi r h \delta x$ $\lim_{\delta x \rightarrow 0} \sum_{x=0}^1 2\pi(x)(3x-x^2) \delta x$ $2\pi \int_0^1 3x^2 - x^3 dx$ $2\pi \left[\frac{3x^3}{3} - \frac{x^4}{4} \right]_0^1$ $2\pi \left[1 - \frac{1}{4} \right] = \frac{3}{2}\pi$	correct use of cylindrical shells method progress solution	
	c) $V = \frac{1}{2} \int_1^2 \pi r^2 dx$ $= \frac{1}{2} \int_1^2 \pi y^2 dx$ $= \frac{\pi}{2} \int_1^2 x^2 - 1 dx$ $= \frac{\pi}{2} \left[\frac{x^3}{3} - x \right]_1^2$ $= \frac{\pi}{2} \left[\frac{8}{3} - 1 \right]$ $= \frac{2\pi}{3}$	correct area set up & progress solution	
	d) $V = \int_0^c \frac{1}{2} \times b \times h dy$ $= \int_0^c \frac{1}{2} \left(\frac{A}{c} y \right) \left(\frac{B}{c} y \right) dy$ $= \frac{1}{2} \int_0^c \frac{AB}{c^2} y^2 dy$ $= \frac{1}{2} \left[\frac{AB}{3c^2} y^3 \right]_0^c$ $= \frac{1}{2} \left[\frac{ABC}{3} - 0 \right] = \frac{ABC}{6}$	correct area set up progress solution	